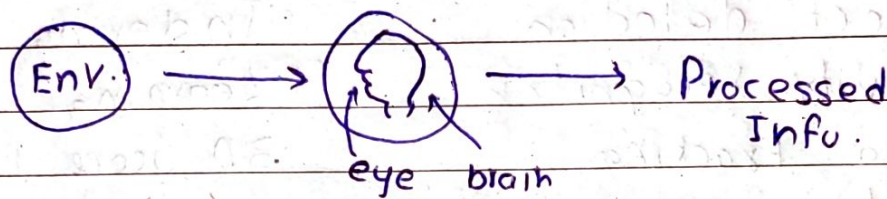


Computer Vision.

- Computer Vision is an interdisciplinary field that deals with how Computers can be made to gain high-level understanding from digital images or videos.
- It Seeks to automate tasks that the human visual System can do



Input → Sensing device → Interpreting device → Output

Computer vision is concerned with the

- Automatic extraction
- Analysis
- Understanding

on useful information from a single or sequence of images.

- Machine Vision refers to a Systems engineering discipline especially in the context of factory automation.

- The image data can take many forms such as
 - Video Sequences
 - Views from multiple cameras
 - Multidimensional data from a medical scanner.

* Classification.

Classification + localization

Object detection

Instance Segmentation

Semantic Segmentation

- box

- All Object box

- Border Segment

- Colour Segment

* Application of CV

• Scene reconstruction

• Object detection

• Activity recognition

• Video tracking

• Motion estimation

• Image restoration

• Indexing

• Learning

• 3D Scene Modeling

* Use Case

• Automatic inspection

• Assisting humans

• Controlling processes

• Tracking

• Lip reading

• Detecting events

• Interaction

• Monitoring

• Navigation

* Image formation

Image formation occurs when a sensor registers radiation that has interacted with physical objects.

* Image Models

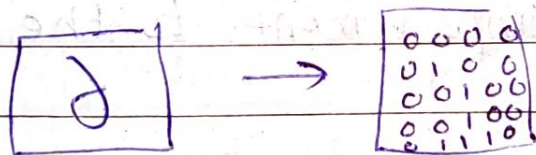
• An image function is the fundamental abstraction of an image

• A geometrical model describes how three d are projected on 2.

- A radiometrical model shows how the imaging geometry, light sources and reflectance properties of objects affect the light measurement at the sensor.
- A Spatial Frequency model describes how spatial variations of the image may be characterized in a transform domain.
- A color model describes how different spectral measurements are related to image colors.
- A digitizing model describes the process of obtaining discrete samples.

* Image functions :

- It is mathematical representation of an image.
- It is a vector-valued function.



- A multispectral image f is a vector-valued function with components $(f_1, f_2, \dots, f_n)$.
- One special multispectral image is a colour image.

$$f(x) = \{f_{\text{red}}(x), f_{\text{blue}}(x), f_{\text{green}}(x)\}$$

* Digital Image:

A special case of the image function is the digital (discrete) image function, where the arguments to and value of the function are all integers.

* Pixel:

Pixel is the smallest addressable element in a raster image.

Or Smallest addressable element in a dot matrix display devices

* In CV input is images output is form of Scalar, Vector or matrix.

* In image processing images are input and output is transformation in some way representing improvement to the original array.

* Image processing deals with improving the pictorial information.

- Improves Contrast

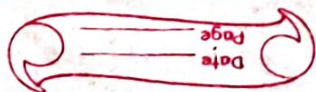
- Removes noise

- Removes blurring

- Corrects geometrical distortions

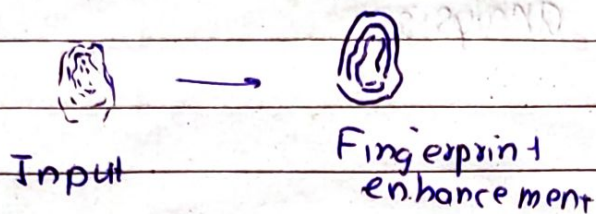
Image Processing $\rightarrow$ Manipulating Img

CV $\rightarrow$ Interpreting Img.



* Low level processing

- It involves image processing tasks in which the quality of the image is improved.
- It helps for human observers and higher level routines.



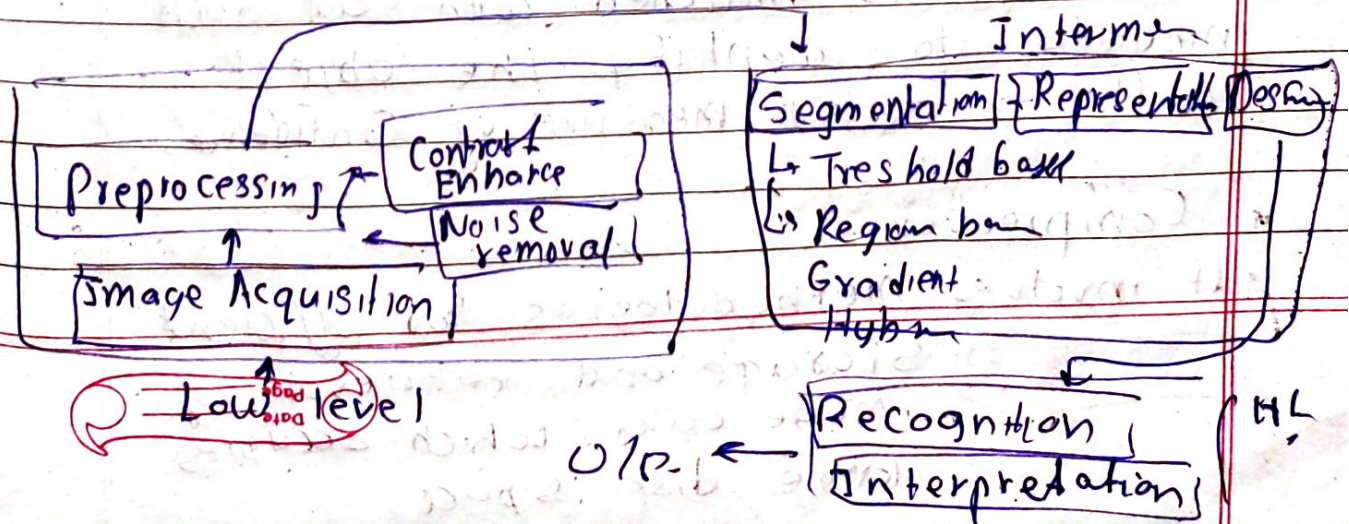
* Intermediate level processing:

- It involves the process of feature extraction and pattern detection task.
- It assist the final tasks of high level vision.

* High level Vision

- It involves autonomous interpretation of scenes, for ex.
 - Classification
 - Recognition
 - Object identification.

- A top down approach, rather than a bottom up is used to design these systems.



There exists four categories of tasks in digital image processing

1. Segmentation
2. Recognition
3. Compression
4. Motion analysis

* Segmentation :

- It deals with the process of fragmenting the image into homogeneous meaningful parts, regions or sub-images
- Techniques used for Segmentation are based on analysis of histogram, edges, colors, textures etc.

* Object Recognition

- It deals with identification or classification of objects in an image
- System is trained using HMM, GMM, ANN
 - HMM - Hidden Markov Model
 - GMM - Gaussian Mixture Model
 - ANN - Artificial Neural Network

Test image is matched with all such models to identify the object with a certain measure of confidence.

* Compression

- It involves methodologies for efficient



Storage and retrieval of image data, which occupies large disk space.

- Image Compression may be
 - Lossy
 - Lossless.

• Lossless compression is preferred where every bit is important. Such as medical applications.

The lossy compression that produces imperceptible differences may be called visually lossless.

* Methods

- Runlength Coding
- Huffman Coding
- Discrete Cosine Transform (DCT) (Lossy)
- Wavelet based (Lossy)

* Motion Analysis :

- Also known as dynamic scene analysis.
- It involves tracking and estimating the path of movement of object/s from a sequence of frames.

• Motion Analysis is based on

- Tracking
- Obtaining correspondence between frames.

* Neighborhood & Connectivity of Pixels.

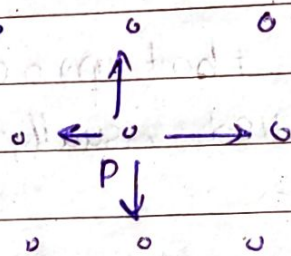
* Neighborhood

N_4 - 4-neighbors (2 horizontal, 2 vertical)

N_D - diagonal neighbors

N_8 - 8 neighbors ($N_4 \cup N_D$)

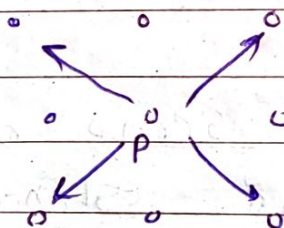
N_4
 $(x+1, y)$
 $(x-1, y)$
 $(x, y+1)$
 $(x, y-1)$



(Each of the neighborhood pixel is at unit distance)

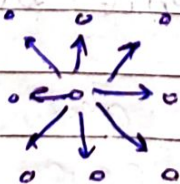
$N_D(p)$: Diagonal Neighbors of Pixel p.

Each neighborhood pixel is at Euclidean distance of $\sqrt{2}$ from p.



$(x+1, y+1)$
 $(x-1, y+1)$
 $(x+1, y-1)$
 $(x-1, y-1)$

N_8



* Adjacency

Two pixels are connected if they are neighbors, and their gray levels satisfy some specific criterion of similarity.

ex: In binary img, two pixels are connected

- > If they are 4 neighbors and
- have same value (0/1)

• Various types of Adjacency

- 4 adjacency
- 8 adjacency
- m adjacency

Let V be set of gray levels used to define adjacency.

4-Adjacency: Two pixel p, q with values from V are 4-adjacent if q is in the set $N_4(p)$

m-adjacency.

IF q is in $N_4(p)$ or
 q is in $N_p(p)$ and set $[N_4(p) \cap N_4(q)]$ is empty.

* Connectivity. ∴ Two pixel p, q that have value from V are

4-Connected, if q is in set $N_4(p)$

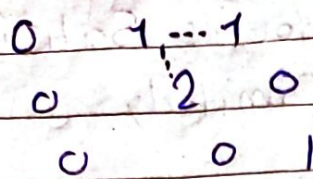
8-Connected, if q is in set $N_8(p)$

m-connected if q is in $N_4(p)$ or

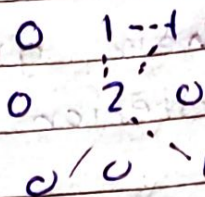
q is in $N_p(p)$ and $N_4(p) \cap N_4(q)$ is empty.

$$V = \{1, 2\}$$

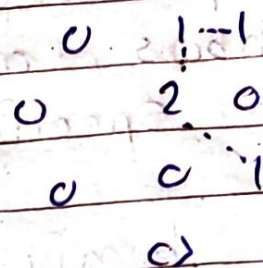
* F_k



a)



b)



c)

* Paths & Path lengths

A path from pixel p with coordinates (x, y) to pixel q with coordinates (s, t) is a sequence of distinct pixels with coordinates:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

where $(x_0, y_0) = (x, y)$

$(x_n, y_n) = (s, t)$

(x_i, y_i) is adjacent to (x_{i-1}, y_{i-1}) $1 \leq i \leq n$

* Connected Components:

• Connected pixels

If p and q are pixels of an img subset S then p is connected to q in S if there is a path p to q consisting entirely of pixels in S .

• Connected Component

For every pixel p in S , the set of pixels in S that are connected to p

is called a connected component of S

- **Connected Set:**
If S has only one connected component then S is called Connected Set.

* Region and Boundaries:

A Subset R of pixels in an image is called a Region of the image in R is a connected set

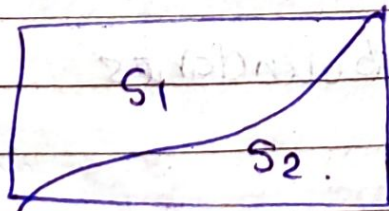
- Each connected component forms a region.

• Boundary of the Region R .

It is the set of pixels in the region that have one or more neighbors that are not in R .

* Adjacency of Image Subsets.

- Pixel p is adjacent to pixel q if they are connected.
- Two image subsets S_1 and S_2 are adjacent if some pixel in S_1 is adjacent to some pixels in S_2 .



* Arithmetic Operations of pixel.

Add, Sub, Mult, Div, AND, OR, Complement

* Operation on pixels based on neighborhood

	P_1	P_2	P_3
	P_4	P_5	P_6
	P_7	P_8	P_9

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$P = w_1 P_1 + w_2 P_2 + \dots + w_9 P_9 = \sum w_i P_i = q$$

* Application of Neighborhood Processing

- Smoothing / Averaging
- Noise Removal / Filtering
- Edge detection
- Contrast enhancement

* Issues in Neighborhood Processing

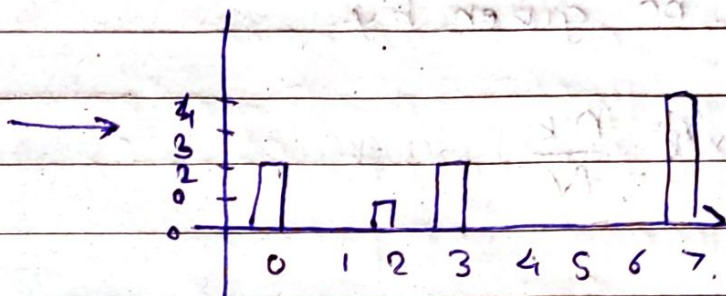
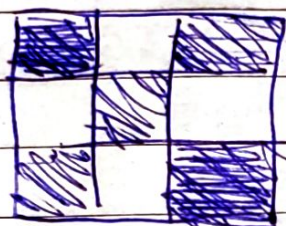
- Choice of w_i (weights)
- Window Size (Choice of N)
- Boundary edge case
 - Pad with zero
 - Pad assuming periodicity
 - Extrapolation
- Do not compute boundaries

* Histogram Processing:

A histogram of an image represents the frequency distribution of pixel intensity values.

X Axis $\rightarrow$ Intensity levels (eg. 0-255)

Y Axis $\rightarrow$ Number of pixels



Black $\rightarrow$ 0

White $\rightarrow$ 7.

$I =$

0	7	2
7	3	7
3	7	0

1-D vector Representation
 $\rightarrow h = [2 | 0 | 1 | 2 | 0 | 0 | 0 | 4]$

I Matrix.

(2D representation)

Notations: In a gray level img, 1

L : No. of gray levels. (0 to $L-1$)

r_k : Normalized intensity val, $r_k \in [0, 1]$

n_k : No. of pixel with gray level r_k

N : Total No. of pixel in img

- Histogram h of an image is defined as

$$h(r_k) = n_k, \quad k = 0, 1, 2, \dots, L-1.$$

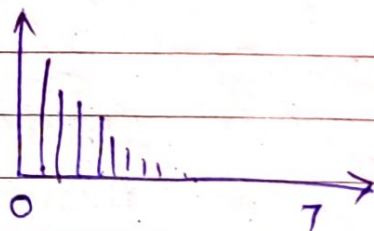
The probability assigned to each gray level can be given by

$$P_r(r_k) = \frac{n_k}{N}$$

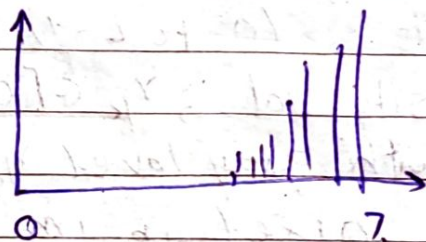
- The plot of $P_r(r_k)$ with respect to r_k is called normalized histogram of the image

- ★ Four basic image types and their histogram.

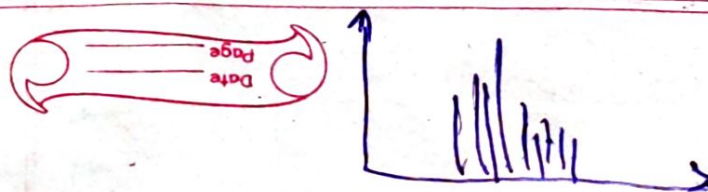
1. Dark



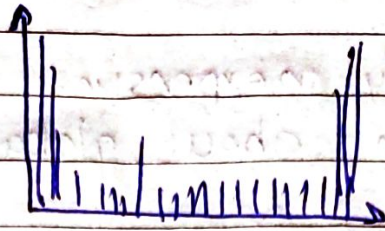
2. Light



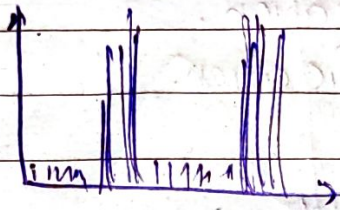
3. Low Contrast



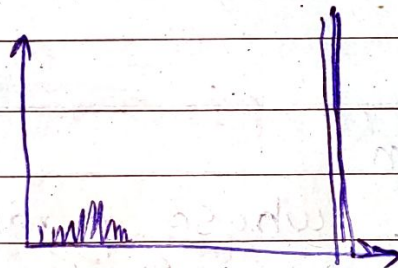
4. high Contrast.



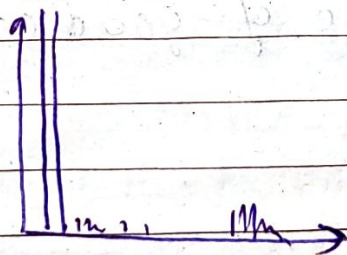
A) Two prominent intensities



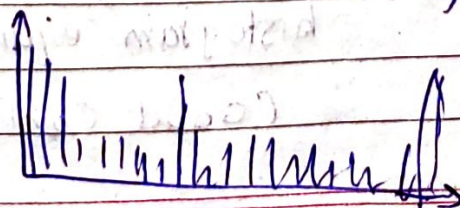
B) Image with white background



c) Image with black background



d). Image with more or less uniform coloring except the background.

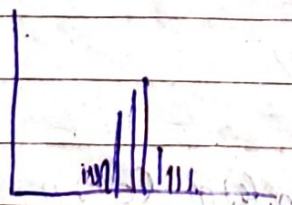


* Applications of Histogram.

1. Simple to Calculate.
 - Computationally inexpensive.
2. Provides information about global Appearance.
 - Brightness
 - Contrast
 - Uniformity.
3. Image Enhancement
 - Histogram Equalization.
 - Histogram Specification.
4. Image Compression
 - Reduce redundancy.
5. Image Segmentation.
 - Separate object from background.

* Good Image histogram

Intuitively images whose pixels occupy the entire range of intensity levels and tend to be distributed uniformly will have an appearance of good contrast.



Input.
(Low Contrast)



histogram equalization
(Good Contrast)

* Histogram Equalization.

• It is a contrast enhancement technique in digital image processing.

• The idea is to redistribute the intensity values of an image so that the output image has a uniform histogram.

• If the histogram of an image is very narrow, the image looks dull (low contrast).

• By spreading the intensities over the full range, the image becomes more visually informative.

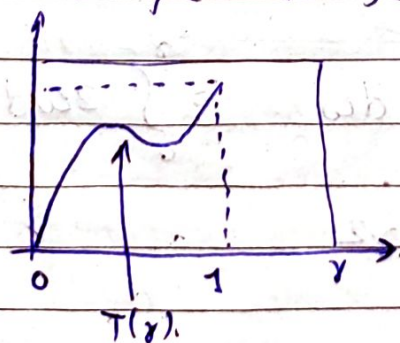
a) Continuous Case:

let x = input pixel intensity. (normalized to $[0, 1]$)

$P_x(x)$ = Probability Density function of x .

$$S = T(x) = \int_0^x P_x(w) dw.$$

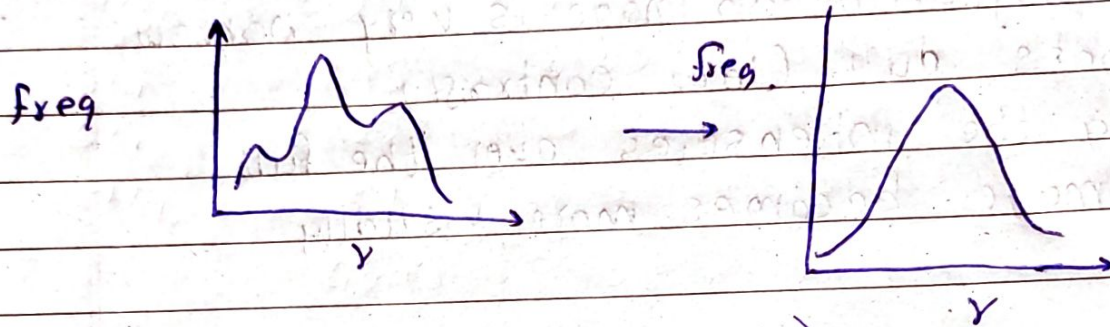
Objective is to get $S = T(x)$ that produces level S for every pixel level x in the original image to get uniform histogram.



Constraint on $T(x)$

$T(x)$ should satisfy following

1. $T(x)$ is monotonically increasing where x is in the range $[0, 1]$
2. $T(x)$ also varies in the range $[0, 1]$



Normalized Histogram (pdf)

Normalized histogram can be considered as a pdf - Gray level in an image can be viewed as random variables in the interval $[0, 1]$

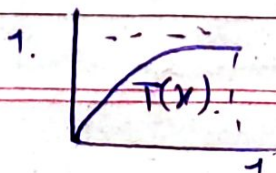
b) Histogram Equalization - Continuous Case

ex. $p_x(x) = 2x$ $0 \leq x \leq 1$
is the pdf

$$S = T(x) = \int_0^x p_x(w) dw = \int_0^x 2w dw$$

$$= [w^2]_0^x = x^2$$

$$S = x^2$$



* Inverse Transformation.

The inverse transformation from S back to r is denoted by $r = T^{-1}(s)$

- To define inverse transformation, $T^{-1}(s)$ should be strictly monotonically increasing function.

i.e.

$$T(r_1) < T(r_2) \text{ if } r_1 < r_2.$$

It ensures one-to-one mapping

- Hence, to get inverse transformation, we need $T(r)$ to be one-to-one and onto (bijective)

b> Histogram Equalization - Discrete Case.

- In discrete case, r takes discrete gray values r_k , $k=0, 1, \dots, L-1$ and
- Probability of occurrence of a gray level r_k in an image is approximated by,

$$P_r(r) = \frac{n_k}{N}$$

$$S_k = T(r_k) = \sum_{j=0}^k P_r(r_j)$$

x_k	n_k	$P_x(x_k)$	$Cdf(S_k)$	Quantizer
0	790	0.19	0.19	1/7
1/7	1023	0.25	0.44	3/7
2/7	850	0.21	0.65	5/7
3/7	656	0.16	0.81	6/7
4/7	329	0.08	0.89	6/7
5/7	245	0.06	0.95	1
6/7	122	0.03	0.98	1
7/7	81	0.02	1.00	1
	<u>4098</u>	<u>1.00</u>		

S_k	n_k	$P_S(S_k)$
0	0	0
1/7	790	0.19
2/7	0	0
3/7	1023	0.25
4/7	0	0
5/7	850	0.21
6/7	985	0.24
1	448	0.11

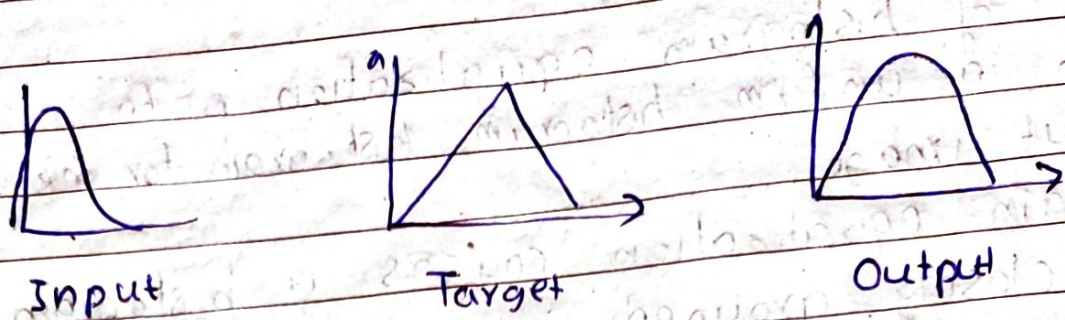
New histogram

* Summary:

- Goal of histogram equalization is to obtain a uniform histogram for the output image.
- Histogram equalization causes a histogram with closely grouped values to spread out into a flat region.
- Spreading or flattening the histogram makes the pixels appear darker and the light pixels appear lighter.
- Histogram equalization is a point process.
- It does not introduce new intensities in the image.
- Existing values will be mapped to new values keeping actual number of intensities in the resulting image equal or less than the original number of intensities.

* Histogram Specification (Matching)

- It is the process of transforming an input image so that its histogram matches a given target.
- It provides gray level transformation such that the histogram of the output image matches to that of the pre specified histogram of a target image.



a) Continuous Case.

Input CDF $S = T(r) = \int_0^r p_r(w) dw$

Desired CDF $V = G(z) = \int_0^z p_z(t) dt$

Since both S and V are uniform $[0,1]$

$$z = G^{-1}(T(r))$$

Steps: Equalize the histogram of input image
 Equalize the specified histogram
 Relate the two equalized histogram

b) Discrete Case.

input CDF $S_k = \sum_{j=0}^k p_r(r_j)$

Desired CDF $V_k = \sum_{j=0}^k p_z(z_j)$

Hence z_k must satisfy $G(z_k) = T(r_k)$

$$z_k = G^{-1}[T(r_k)]$$

ex.

r_k	n_k	$P_k(r_k)$	z_k	$P_z(z_k)$
0/7	790	0.19	0	0
1/7	1023	0.25	1/7	0
2/7	850	0.21	2/7	0
3/7	656	0.16	3/7	0.15
4/7	329	0.08	4/7	0.20
5/7	245	0.06	5/7	0.30
6/7	122	0.03	6/7	0.20
7/7	81	0.02	7/7	0.15

Input Histogram

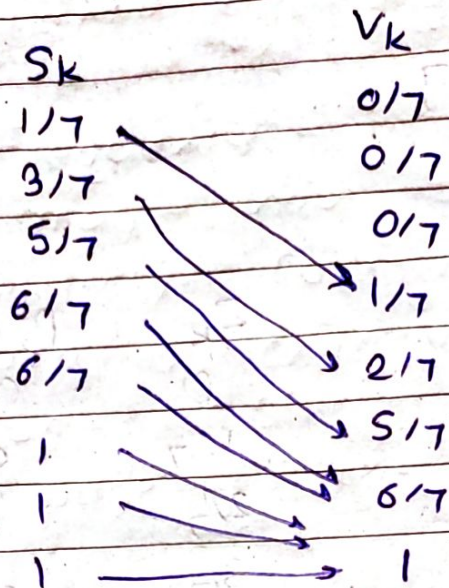
Target Histogram

r_k	P_k	Cdf(P_k)	Gray level s_k
0/7	0.19	0.19	1/7
1/7	0.25	0.44	3/7
2/7	0.21	0.65	5/7
3/7	0.16	0.81	6/7
4/7	0.08	0.89	6/7
5/7	0.06	0.95	1
6/7	0.03	0.98	1
7/7	0.02	1.00	1

equalization of second histogram

z_k	$P(z_k)$	Cdf	Gray level u_k
0	0	0	0/7
1/7	0	0	0/7
2/7	0.15	0.15	0/7
3/7	0.20	0.35	1/7
4/7	0.30	0.65	2/7
5/7	0.20	0.85	5/7
6/7	0.15	1.00	6/7

Map



Mapping function

1	4
2	5
3, 4	6
5, 6, 7	7

Output

z_k	n_k	$P_z(z_k)$
0	0	0
1/7	0	0
2/7	0	0
3/7	798	0.19
4/5	1023	0.25
5/7	850	0.21
6/7	985	0.24
7/7	448	0.11

* Fourier Transform:

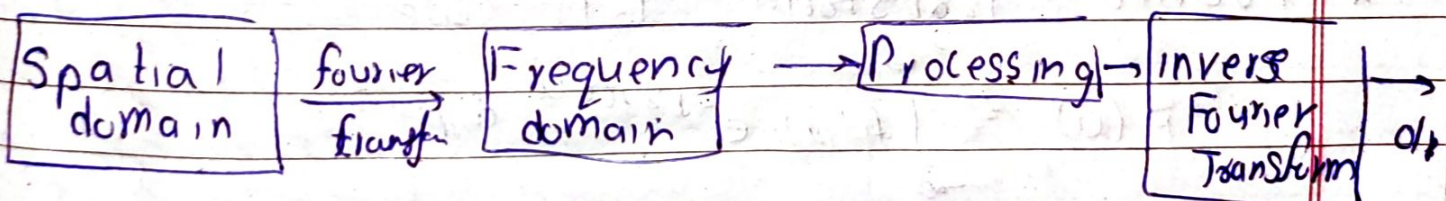
The Fourier transform states that a non periodic function whose area under curve is finite can also be represented as weighted sum of sin and cosine functions of varying frequency.

- It is a mathematical tool that transforms a signal from the time / Space domain into the frequency domain.

Many real world signals (audio, image, video) are time - domain or spatial - domain.

We need Fourier transform as

- Better analysis and interpretation.
- Some of the operations can be efficiently performed in Fourier domain.



* Inverse Fourier transform

tool which turns a Fourier space description of an image back into real space



* Significance of Frequency

1. High Frequency:

If an image has large values of high frequency components then it means the data is changing rapidly on a short distance scale.
ex. a page of text

~~ABCDEFGHIJ~~ →

2. Low Frequency:

- If an image has large low frequency components then it means the large scale (slow varying) features of the picture are more prominent
- eg. a single simple object which occupies most of the image.

* Fourier Transform 1-D Case

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi xu} dx$$

$f(t)$ = original signal

$F(u)$ = Fourier Transform.

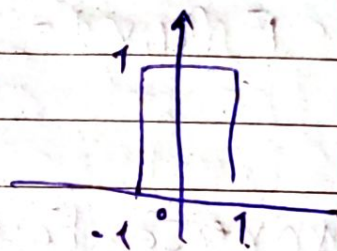
$F(u)$ will be complex quantity even when $f(x)$ is real.

* Inverse Fourier Transform

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi xu} du$$

It is similar to Fourier transform except that the exponential term has opposite sign

ex. $f(x) = \begin{cases} 1 & \text{if } |x| \leq 1 \\ 0 & \text{otherwise} \end{cases}$



$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi xu} dx$$

$$= \int_{-1}^1 e^{-j2\pi xu} dx$$

$$= \frac{-1}{j2\pi u} (e^{-j2\pi u} - e^{j2\pi u})$$

$$= \frac{\sin 2\pi u}{\pi u}$$

ex. $f(x) = \begin{cases} A & 0 < x < x \end{cases}$

$$F(u) = \int_0^x A e^{-j2\pi xu} du$$

$$= \frac{-A}{j2\pi u} [e^{-j2\pi xu}]_0^x$$

$$= \frac{-A}{j2\pi u} (e^{-j2\pi ux} - 1)$$

$$= \frac{A}{\pi u} \sin(\pi ux) / x$$

* Magnitude & Phase

$$F(u) = R(u) + jI(u)$$

$$\text{Power Spectrum} = P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$

* Fourier Transform 2-D Case

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(xu + yv)} dx dy$$

* Inverse Fourier transform

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(xu + yv)} du dv$$

* Discrete Fourier Transform:

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-j2\pi xu/N}$$

$$f(x) = \sum_{u=0}^{N-1} F(u) e^{j2\pi xu/N}$$

- It is used for digital signals / images
- We replace integration by summation.

$F(u)$ and $f(x)$ makes discrete Fourier transform pair

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} \left[\sum_{u=0}^{N-1} f(x) e^{j2\pi x u / N} \right] e^{-j2\pi x u / N}$$

$$\Rightarrow F(u) \equiv F(u)$$

* Discrete Fourier transform 2D case

$$F(u, v) = \frac{1}{NM} \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f(x, y) e^{-j2\pi(xu/N + yv/M)}$$

$$f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{M-1} F(u, v) e^{j2\pi(xu/N + yv/M)}$$

When Image Size is $N \times N$
then $M = N$

$$F(u, v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{xu + yv}{N} \right)}$$

$$f(x, y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{xu + yv}{N} \right)}$$

* $F(0, 0)$ represents DC-Component of the image which corresponds to the average brightness.

* $F(N-1, N-1)$ represent the highest frequency



* Visualization of DFT.

• The Fourier transform produces a complex valued output image which can be displayed with two images

- either with real and imaginary part
- with magnitude and phase.

In image processing only magnitude is displayed, as it contains most of the information of the geometric structure of the spatial domain image

• Since dynamic range of $|F(u,v)|$ is high stretching is applied

$$D(u,v) = \log(1 + |F(u,v)|)$$

• To be able to convert image back to spatial. values are stored in float as Fourier domain has high range.

• Fourier transform Spectrum (Magnitude) image is always Symmetrical about the center.

The logarithmic scaling makes it difficult to tell the influence of single frequency in the original image.

* Fast Fourier Transform

- As the result of Fourier transform is Symmetric about mean.
- So FFT generates only the left half of the image
- The other half image is generated by rotation and duplication of generated data
- Also while converting back to Spatial it only sees left half and ignores right half.

* Magnitude Vs Phase

- Magnitude determines the relative presence of a Sinusoid.
- Phase determines the relative phase (placement) of the Sine and Cos wave

Q Which is more important?

→ Phase.

Does one component contains more info than other

→ Yes, phase.

* Complexity of 1-D Fourier transform

N multiplication, $N-1$ addition ($2N-1$) operations.

$$N(2N-1) = 2N^2 - N$$

$$= \underline{\underline{O(N^2)}}$$



Using FFT this is reduced to $O(N \log(N))$

* 2D Fourier Transform
→ $O(N^4)$

* Computational Complexity of 2D reduction.

- Spatial domain converted to intermediate using N 1-D Fourier trans (rows)

- Intermediate is transformed to final again using N one d mens. (col) transform

reduces to → $O(N^3)$

or $O(N^2 \log N)$ using FFT

* Performing filtering in frequency domain

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{otherwise} \end{cases} \quad \left. \vphantom{\begin{matrix} 1 \\ 0 \end{matrix}} \right\} \begin{array}{l} \text{low pass} \\ \text{filtering} \\ \text{(Band pass)} \end{array}$$

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{otherwise} \end{cases} \quad \left. \vphantom{\begin{matrix} 0 \\ 1 \end{matrix}} \right\} \begin{array}{l} \text{High Pass} \\ \text{filtering} \\ \text{(Band stop)} \end{array}$$

* Properties of DFT.

• Conjugate Symmetry.

$$|F(u,v)| = |F(-u,-v)|$$

- Fourier transfer of real even function is real
- Fourier transfer of real odd function is imaginary

* Properties of Fourier transform

1. Scaling

$$a f(x,y) = a F(u,v)$$

2. Distribution

$$F[f(x,y) + g(x,y)] = F[f(x,y)] + F[g(x,y)]$$

But for product $\neq$

3. Rotation.

Rotating $f(x,y)$ by θ rotates $F(u,v)$ by θ .

4. Average Value

$$\bar{f}(x,y) = \frac{1}{N} F(0,0)$$

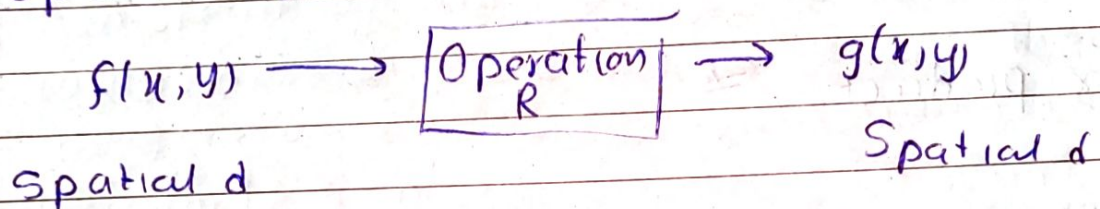
* Image Filtering:

- The removal of unwanted signal or undesirable characteristics from the image
- It is used to enhance or modify the image

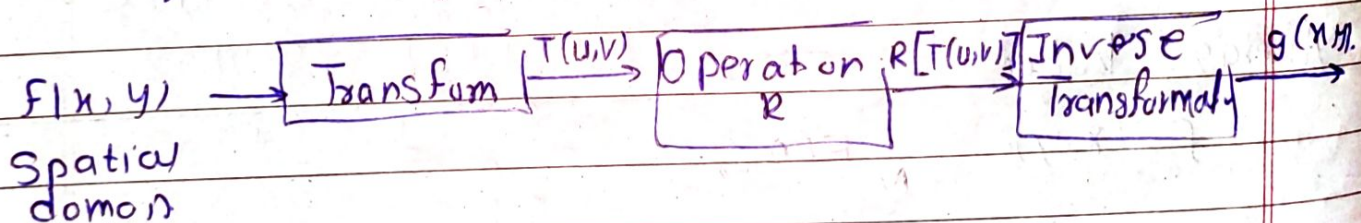
$$f(x,y) \xrightarrow{\text{filtering}} g(x,y)$$

- Filtering basically comes from frequency domain

• Spatial Domain

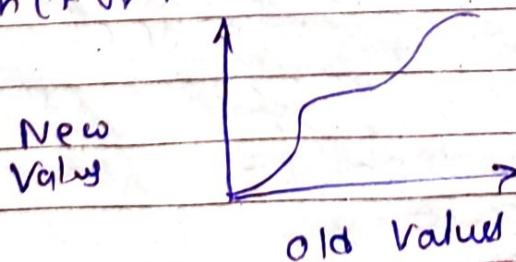


• Frequency domain



* Point processing Methods

Converting a given pixel value to a new pixel value based on some predefined function.



Ex. Negative

Contrast stretching

Thresholding

Histogram Equalization



$$g(x,y) = T(f(x,y))$$

$f(x,y)$ = Input pixel intensity

• No neighborhood (like Convolution) is used.

* Area or Mask Processing Methods.

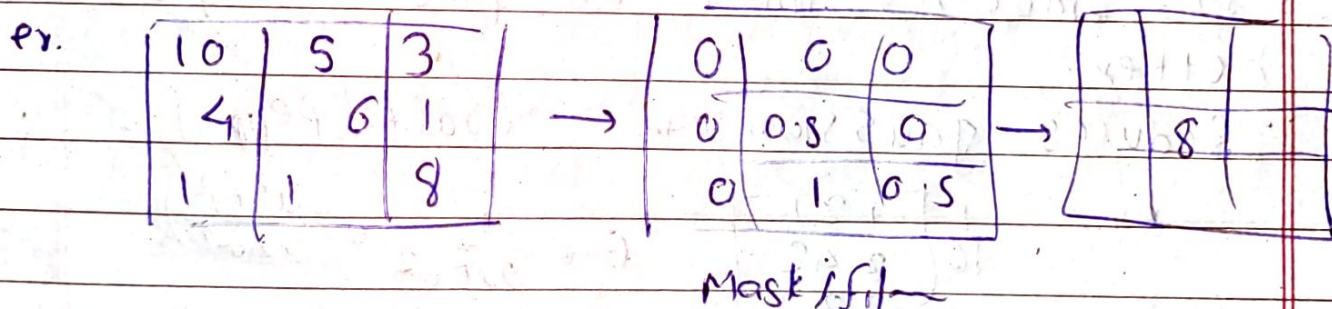
• Output pixel $g(x, y)$ depends on input pixel $f(x, y)$ and its neighbors.

• A mask (kernel, filter, window) is placed over image and a weighted operation is performed.

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

Area shape is defined using mask.

ex. Smoothing, Sharpening, edge detection



* Linear Filtering : Uses weighted sum.

• Correlation

• Convolution

Correlation : A filter image is generated as the center of the mask visits every pixel in the input image.

Handling Boundary — 1. 0 Padding
2. Wrap around.

* Convolution

Same as correlation, but with kernel rotated by 180° .

* Average Mean (Box) Filter.

- Each output pixel is the average of its neighbors in a defined mask.

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

* Gaussian Filter.

- Instead of uniform weights, it uses a gaussian distribution to assign higher weight to the center pixels and lower to farther pixels.
- It smooths image but preserves edges better.
- Reduces gaussian blur (Salt & Pepper) -

ex.

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

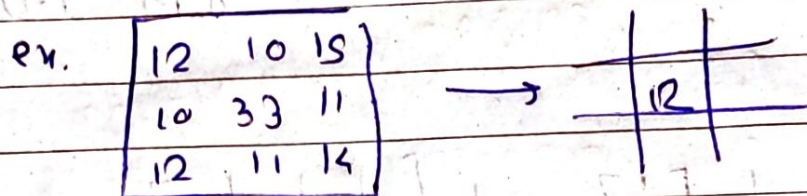
$$G = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Non Linear Filtering: Doesn't rely on linear sums
ex Median filter, Max/Min filters.

* Application of Smoothing: Noise Removal.

* Median Filter :

In this the center pixel in kernel is replaced by the median pixel in neighborhood.



Adv : Occasional Spurious high or low value is not averaged in

The sharpness is preserved.

* Rank Order Filters.

Rank all pixel values in a window

- Take nth order value

ex Max or min or median.

Order Statistic Filters.

• These are non linear Spatial filters.

• They work by ordering (sorting) pixel values within neighborhood

ex. Median, Min, Max.

Area

→ Linear

↳ Smoothing

↳ Sharpening

→ Avg, Gauss

→ Laplacian, Gradient

→ Non Linear

↳ Median

↳ Max Min



Hybrid Filters :

- Combination of
Convolution
Order Statistics.